

Gideon A. Ong

Mechanical Engineering Student · English/Spanish Bilingual

ga09819@utulsa.edu | +1 (918) 409-2444 | gideonong.me | linkedin.com/in/gideon-a-ong

Education

The University of Tulsa, Tulsa, OK

In progress

International Engineering and Language Program (IEL)

Mechanical Engineering, B.S.M.E.; Spanish, B.A.; International Business, B.S.B.A.

GPA: 4.0

Skills

Languages: English (fluent), Spanish (fluent), French (beginner)

Programming: Python (pandas, matplotlib, scikit-learn, Tkinter, pytest, pyautogui), MATLAB, Java, R,

JavaScript/Google Apps Script, Mathematica, Stata, Arduino

Engineering and data software: SolidWorks (including Simulation), PolyWorks, QGIS, Excel, Google Sheets, GitHub, GitHub Copilot, Inno Setup

Design and media: GIMP, Inkscape, Canva, Figma, Publisher, PowerPoint, Audacity, ProPresenter

Machine shop: CNC and manual mills and lathes, MIG welding, saw, sander, 3D printing

Certifications: CSWP (Certified SolidWorks Professional), GitHub Copilot, Certiport Java Specialist, McGraw-Hill

Excel Black Belt – Level 4

Experience

Engineering Intern, ALS R&D · Baker Hughes, Claremore, OK

Summer 2026

- Co-built Velora with Zohaib Sheikh: a Python app that turns dyno test data into finished reports, cutting 8–40 hour report times by 80%+ (worth an estimated \$200k a year).
- Built AutoScan, a Python app that automates a KEYENCE 3D optical scanner so scan batches run unattended, saving about 10 hours of operator time per batch (at least \$5k a year in labor).
- Spun off G-View, a standalone data-review app, and G-RPA, a no-code automation builder, from those two projects; also built G-Transcript, a tool for editing and exporting AI-generated meeting transcripts.
- Wrote user and technical guides for each app with AI assistance, cutting documentation time from weeks to days.
- Presented my GitHub Copilot workflow to the R&D team, gave peer-to-peer training, and wrote lessons-learned documents; was asked to write a white paper on the workflow.
- Reviewed and revised drawings for a precision stage tester fix, using 3D-printed parts to check fit before machining.
- Designed and 3D-printed parts including scan fixtures, chuck spacers, a smaller ultrasonic-wash basket, and a splash guard for a visual flow test loop; iterating on printed prototypes saved at least 2 weeks of my design and lead time.

Machine Shop Technician · McElroy Prototyping Lab, The University of Tulsa

Spring 2026 – present

- Machine parts on CNC and manual mills and lathes, mostly aluminum for student projects, plus some plastics and other materials for student clubs and research groups.
- Help students use the shop safely and effectively.
- Machined the valve-actuating pin for my team's prize-winning pump cylinder failure analysis.

Grader & Data Analyst · Dr. Joshua Schultz, The University of Tulsa

Spring 2026 – present

- Grade homework for Intro to Dynamics and give Dr. Schultz quick charts and summary statistics for each assignment.
- Built a Python app that merges each student's scans, PDFs, Word files, and code into one PDF per student for grading on an iPad.

Undergraduate Research Assistant, TURC · Dr. Janica Edmonds, The University of Tulsa *Summer 2025*

- Designed a Math Teachers' Circle session on map projections, a topic new to the circle, and led it with Tulsa-area teachers in October 2025.
- Reviewed research on math circles and teacher development to suggest ways the Tulsa Math Teachers' Circle could improve its communication and session structure; drafted further session ideas on circle inversion and linguistics puzzles.
- Presented the project on a poster at a TU research event.

Projects

G-View: Test Data Review App · Baker Hughes *Summer 2026*

- As sole developer, built a desktop app for reviewing test-data spreadsheets with hundreds of thousands of rows: plot the data, crop out bad sections without touching the source file, and export clean copies.
- Built with Python, pandas, matplotlib, Tkinter, and scikit-learn clustering; shipped with an Inno Setup installer and user and technical guides.

AutoScan: 3D Scanner Automation · Baker Hughes *Summer 2026*

- Added PolyWorks integration for go/no-go analysis, custom scan recipes, a no-code maintenance screen, and an Inno Setup installer.

Paint Sprayer Pump Cylinder Failure Analysis · ME 3033 Properties of Materials *Spring 2026*

- Traced cracked pump cylinders on a client's paint sprayers (\$10–15k per failure) to rinse water freezing inside, using hardness testing, microstructure analysis, and thick-wall stress analysis.
- Served as technical lead for the prototype (overall team lead: Rachel Lewis); co-designed and machined a spring-loaded pin that lifts the valve so the cylinder drains in about 20 seconds, as a \$135 retrofit.
- Won the 2026 Hackworth-Willson Prize for Excellence in Failure Analysis.

Keplinger Hall Heating Analysis · ES 3073 Heat Transfer *Spring 2026*

- Estimated winter heat loss for TU's engineering building and sized a hot-water radiator system for it, using Python and Excel.

Finding Notes in Music with MATLAB · MATH 5353 Discrete/Integral Transforms *Spring 2026*

- Wrote a MATLAB script that finds the main notes and overtones in a song using windowed Fourier transforms.

Gender and Employment in Computer Science · ECON 4273 Applied Econometrics *Spring 2026*

- Used linear probability and logistic models in Stata to test whether gender affects the odds of being employed in computer science jobs.

Stress Relief for a Notched Beam · ES 3023 Mechanics of Materials *Fall 2025*

- Used SolidWorks simulation to design a spline cutout that reduced a notched beam's stress concentration factor by 42% (1.39 vs. 2.38) in four-point bending tests.

Chilled Water Pipeline Design · ES 3003 Fluid Mechanics *Fall 2025*

- Routed and sized a replacement chilled-water pipeline for TU's campus in Python, using AI-assisted convex-hull routing and a separate pipe-diameter optimization; it scored 100% with the lowest-cost bid in the class.

Diesel vs. Dual Cycle Comparison · ES 3053 Thermodynamics *Fall 2025*

- Modeled Diesel and dual combustion cycles in Python with NASA property data to compare efficiency, cost, and peak temperature.

Axial Mechanical Fuse · ES 3023 Mechanics of Materials *Fall 2025*

- Designed a notch for an acrylic link to break in the middle at the highest possible load, using a Java search program; two of three test links broke at the notch.

Recurring Crash Hotspots in the UK · ES 4863 AI for Engineers *Spring 2025*

- Found UK intersections that stayed crash hotspots year after year by running DBSCAN on 1.6 million accident records (2005–2014), mapped them in QGIS, and confirmed them against local news reports.

Predicting Bicycle Crash Severity · ES 4863 AI for Engineers

Spring 2025

- Tested FAMD and undersampling to predict UK bicycle crash severity (about 40,000 crashes, 70 features), using SHAP to trace results back to the original variables.

Classifying Landmines from Passive Sensor Data · ES 4863 AI for Engineers

Spring 2025

- Compared decision trees and neural networks, tuned with grid search, to classify five landmine types from UCI passive sensor data; defined a "deadliness" metric for mines misclassified as no mine.

Comparing Crash Severity Models in R · QM 2023 Statistics II

Fall 2024

- Compared naive Bayes, logistic regression, and classification tree models to predict crash severity on a California dataset of 112,660 crashes; presented them in an interactive, parameterized R Markdown slideshow.

Arduino Obstacle-Course Robot · ME 1123 Design & Manufacturing

Fall 2024

- Designed and coded the Arduino control system for a three-person team's robot that carried a ball through an obstacle course.

Leadership & Service

Eagle Scout Project: Wheelchair-Accessible Pathway · The Bee Community, Bryan, TX

Summer 2022

- Planned and led 27 volunteers (about 285 total hours) to build a 3-ft-wide, 273.5 sq ft ADA-compliant decomposed granite pathway for a business that employs adults with special needs.
- Met with the business to define the need, had local experts review the design against ADA guidelines, and completed the build in one nine-hour day.

Honors & Awards

Hackworth-Willson Prize for Excellence in Failure Analysis (2026)

National Varsity Champion – United States Geography Championships (2023, 2024)

Silver Medalist – International Geography Olympiad (2023, 2024)

National Merit Finalist (2024)

Eagle Scout (2023)